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Original Research

Impact of policy response on health protection and economic recovery in OECD and BRIICS countries during the early stages of the COVID-19 pandemic

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ABSTRACT

Objectives: During the early stages of the COVID-19 pandemic, the full reopening of the economy typically accelerated viral transmission. This study aims to determine whether policy response could contribute to the dual objective of both reducing the spread of the epidemic and revitalising economic activities.

Study design: This is a longitudinal study of Organization for Economic Cooperation and Development (OECD) and Brazil, Russia, India, Indonesia, China, and South Africa (BRIICS) from the first quarter (Q1) of 2020 to the same period of 2021.

Methods: From a health-economic perspective, this study established a framework to illustrate the following outcomes: suppression-prosperity, outbreak-stagnancy, outbreak-prosperity and suppression-stagnancy scenarios. Multinomial logistic models were used to analyse the associations between policy response with both the pandemic and the economy. The study further examined two subtypes of policy response, stringency/health measures and economic support measures, separately. The probabilities of the different scenarios were estimated.

Results: Economic prosperity and epidemic suppression were significantly associated with policy response. The effects of policy response on health-economic scenarios took the form of inverse U-shapes with the increase in intensity. 'Leptokurtic', 'bimodal' and 'long-tailed' curves demonstrated the estimated possibilities of suppression-prosperity, outbreak-prosperity and suppression-stagnancy scenarios, respectively. In addition, stringency/health policies followed the inverted U-shaped pattern, whereas economic support policies showed a linear pattern.

Conclusions: It was possible to achieve the dual objective of economic growth and epidemic control simultaneously, and the effects of policy response were shaped like an inverse U. These findings provide a new perspective for balancing the economy with public health during the early stages of the pandemic.

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Introduction

The COVID-19 pandemic has caused the most severe damage to the socio-economic system in the last 80 years and has led to global recession and intense conflicts.[1] Over the last 2 years, governments around the world have been trying to revitalise the socio-economic structure.[2] However, the full reopening and recovery of the socio-economic system relied on the mobility of people to consume and produce, which accelerated the spread of the virus, especially during the early stages of the outbreak.[3–5] Thus, the pandemic and the economy seemed to be coupled and bound together under the bridging effects of mobility, resulting in conflicts between protecting public health and revitalising the economy at the same time.[6–8]

Previous studies generally agreed that both the needs of public health and the economy were important,[9,10] but the logistics for achieving these two goals were different. Some studies demonstrated that only when the epidemic was under control could the economy develop.[11] It was highlighted that 'active' policy interventions were useful to curb transmission, reduce mortality, prevent medical systems from being overwhelmed and promote safe and sustained recovery of the socio-economic system.[12–16]

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Other studies stressed that society had the capacity for epidemic prevention only when sustainable economic growth was maintained.[17] These studies suggested that an open and liberal approach was required, where governments should assume a 'passive' role and phase out segregation, virus testing and vaccination orders to avoid substantial economic damage.[17–21]

All these studies, both those considering epidemic control before economic revitalisation or the reverse, still regarded the pandemic and the economy as absolute antagonists in the short term. Moreover, both 'active' and 'passive' responses, based on this antagonistic relationship, provided limited suggestions for the dual objective of developing the economy and controlling viral transmission concurrently. For this reason, this study defined the economy and the health as a coupled/decoupled system to eliminate the antagonistic perspective. Instead of debating 'active' or 'passive', this study analysed what level of intensity and what kind of policy response could decouple economic recovery from pandemics and achieve the dual objective.

Accordingly, 38 OECD (Organization for Economic Cooperation and Development) and BRIICS (Brazil, Russia, India, Indonesia, China and South Africa) countries, representing the major global economies, were included in this study. First, a four-quadrant framework was provided to depict four scenarios addressing both the economy and the pandemic. Second, it was found that the effects of policy response were lagged and showed an inverted U-shape, and the probabilities of the scenarios of interest corresponded to three different U-shaped curves, which could be described as 'leptokurtic', 'bimodal' and 'long-tailed'. Moreover, 20 policies were classified into stringency/health measures or economic support measures, and heterogeneity analyses took place.

Methods

Data from Q1 2020 to Q1 2021 were used in the analyses. Data came from OECD and BRIICS countries (see Table S1 in the supplementary material). The longitudinal design of the study and the lagged outcomes strengthened confidence in making causal inference.

Outcomes

A framework was established with decoupling theory and defined outcomes[22] (see Table S2 in the supplementary material). The ratio between the relative economic recovery index (ΔERI) and the relative number of new cases (ΔNC) was used to describe the decoupling relationship between the economy and the pandemic (see Equation S1 and Table S2 in the supplementary material). In particular, the ΔERI could be obtained by global principal component analysis of seven core economic indicators (e.g. the change rate of gross domestic product and the change rate of purchase management index on year-on-year basis).[23–25] Quarters in 2019 were set as the common base periods, and data details can be found in Tables S3 and S4 in the supplementary material. The relative number of new cases referred to the difference between the number of newly reported cases per million in each nation and worldwide over a specific period,[26,27] which was available in current data series and could serve directly as the indicator for cross-country comparisons.[28] Pandemic data were collected from the World Health Organisation database.

Four logical relationships, calculated as positive decoupled (ΔERI > 0, ΔNC < 0), negative decoupled (ΔERI < 0, ΔNC > 0), positive coupled (ΔERI > 0, ΔNC > 0) and negative coupled (ΔERI < 0, ΔNC < 0), were used to demonstrate the following four health-economic scenarios: suppression-prosperity, outbreak-stagnancy, outbreak-prosperity and suppression-stagnancy (Fig. S1 in the supplementary material). The suppression-prosperity scenario was the optimal state, where public health and economic development were both promoted. The outbreak-prosperity and suppression-stagnancy scenarios were suboptimal conditions. In the outbreak-prosperity scenario, the economy kept growing while the virus swept over the country. Conversely, suppression-stagnancy referred to the scenario where the proportion of confirmed cases was lower than in other countries, but the economy was seriously damaged. Both scenarios were temporary, and occasionally, they might transition into one another. Finally, the rising number of confirmed cases and sluggish economy appeared at the same time in the outbreak-stagnancy scenario. All outcomes were lagged to establish temporality.

Independent variables

The core explanatory variable, policy response index (PRI), was made up of 20 policies to measure the strength of the government's response.[29] These data were drawn from the Oxford COVID-19 Government Response Tracker (OxCGRT; see Table S5 in the supplementary material). In addition, public health measures and economic measures were assessed using the stringency/health policy index (SHI) and the economic support policy index (ESI), respectively. These indicators were set as the average of daily indices in the quarter.

National and digital features were adjusted for in the data, as previous studies reported this to be crucial for economic development models during the epidemic.[30–34] Adjusted covariates included the population, digital services and online education. The digital services came from the search quantification points with 'Amazon', 'eBay', 'Taobao' and 'Tmall' as the keywords in Google trends; online education was from the search quantification points with 'Coursera', 'Google Classroom', 'Udemy' and 'Zoom'; and population data were derived from the OECD database. Descriptive statistics for all variables are shown in Table S6 in the supplementary material.

Statistical analyses

Multinomial logistic models with random effects were used to analyse the impact of policy response. The standard form of multinomial logistic model was as follows (equation (1)):

$$\ln\left(\frac{\pi_{ij}}{\pi_{ij^*}}\right) = \alpha_j + \beta_j x \quad (j \neq j^*) \tag{1}$$

where π_{ij} was the probability that sample i fell into result j, and j* was the base category; each result had its own slope item; $\frac{\pi_{ij}}{\pi_{ij^*}}$ was known as relative risk. In this study, the dependent variable included four scenarios: suppression-prosperity, outbreak-prosperity, suppression-stagnancy and outbreak-stagnancy. Outbreak-stagnancy scenario was set as the base category. Due to the correlations between data within each sample, random effects calculations were used on the assumption that different nations had unique intercept terms in the regression, which could account for depended observations.

Overall PRI and its subset SHI and ESI, as core independent variables, were placed in the regression models. Models with squared terms of the variable indicated parabolic impacts of the policy response, whereas models without squared terms indicated linear effects. Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) were used to determine whether to include squared terms. The policy mix of stringency/health measures and economic support measures were also analysed.

When estimating the probability of each health-economic scenario, equation (1) could be written as follows:

$$P(y_i = j|x_i) = \frac{\exp\left(x_i'\beta_j\right)}{\sum_{k=1}^{J}\exp\left(x_i'\beta_k\right)} \tag{2}$$

Obviously,

$$\sum_{k=1}^{J}\exp\left(x_i'\beta_k\right) = 1 \tag{3}$$

The coefficient of the base category was set as 0. Thus, the probability of scenario j was as follows:

$$P(y_i = j|x_i) = \begin{cases} \dfrac{1}{1+\sum_{k=2}^{J}\exp(x_i'\beta_k)} & (j=1) \\ \dfrac{\exp(x_i'\beta_j)}{1+\sum_{k=2}^{J}\exp(x_i'\beta_k)} & (j=2,\ldots,J) \end{cases} \tag{4}$$

where the base category, outbreak-stagnancy scenario, was corresponding to j = 1.

Adjusted variables and models were used for sensitivity analyses. First, fixed effects models were used for comparison with the main analyses and Hausman tests were performed. Second, the outbreak-stagnancy scenario was set as the base category (x = 0) and the others as 1; and, separately, the suppression-prosperity scenario was also set as the base category and the others as 1. A series of logistic regressions was performed. STATA (version 17) was used for statistical analyses.

Results

Health-economic scenarios and coupled correlation during the early stage of the pandemic

Fig. 1 shows the coupled relationship between economic expansion and virus spread caused by the outbreak. Many countries entered a paradigm of suppression-stagnancy or even outbreak-stagnancy by Q2 2020 (Fig. 1A). Data from Q3 2020 indicated that a recovery of the health-economic system might be imminent (Fig. 1B). However, only a small number of nations, such as South Korea and China, remained in the suppression-prosperity scenario in Q4 2020 and Q1 2021 (Fig. 1C and D). On the other hand, for example, the United States experienced a high rate of economic

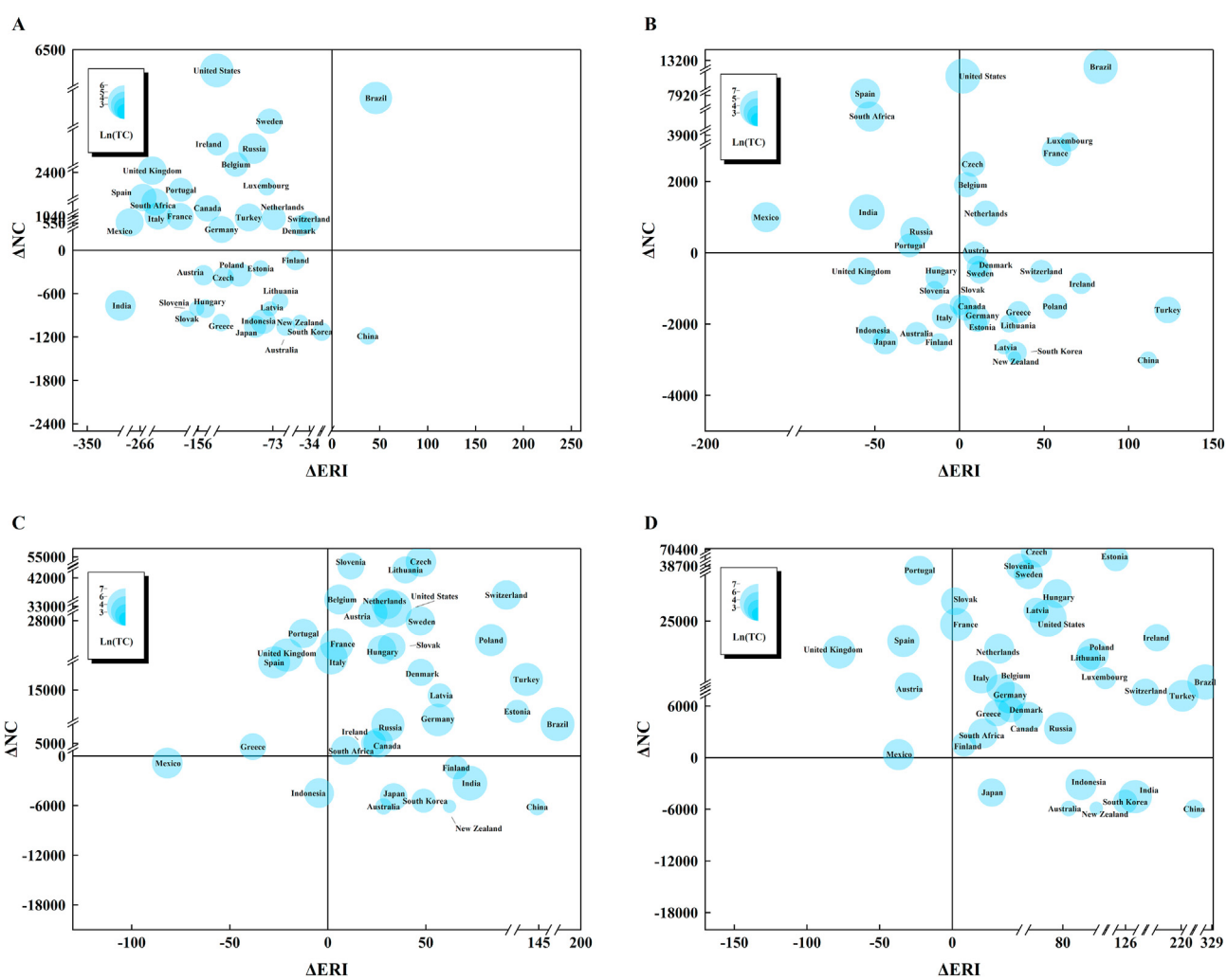

Fig. 1. A schematic framework of the health-economic scenarios in Q2 2020 – Q1 2021 (A–D). Outbreak-prosperity scenario is the first quadrant, outbreak-stagnancy scenario is the second quadrant, suppression-stagnancy scenario is the third quadrant, and suppression-prosperity scenario is the fourth quadrant. ΔERI, relative economic recovery index; ΔNC, relative number of new cases.

Table 1
Results for multinomial logistic models.

Scenario		Model Total	(For reference)	Model Subtype-stringency/ health policies	(For reference)	Model Subtype-economic-policies	(For reference)	Model Subtype
		β (95% CI)	β (95% CI)	β (95% CI)	β (95% CI)	β (95% CI)	β (95% CI)	β (95% CI)
Outbreak-stagnancy scenario	Base category	/	/	/	/	/	/	/
Suppression-prosperity scenario	L_PRI	**0.970***** **(0.271, 1.668)**	0.082*** (0.028, 0.136)	/ /	/ /	/ /	/ /	/ /
	L_PRI_squ	**−0.009***** **(−0.015, −0.003)**	/ /	/ /	/ /	/ /	/ /	/ /
	L-SHI	/ /	/ /	0.750*** (0.271, 1.229)	0.073*** (0.020, 0.126)	/ /	/ /	**0.799***** **(0.280, 1.318)**
	L-SHI_squ	/ /	/ /	−0.007*** (−0.012, −0.002)	/ /	/ /	/ /	**−0.008***** **(−0.012, −0.003)**
	L-ESI	/ /	/ /	/ /	/ /	0.092* (−0.014, 0.198)	0.027* (−0.002, 0.056)	**0.021** **(−0.013, 0.054)**
	L-ESI_squ	/ /	/ /	/ /	/ /	−0.001 (−0.001, 0.0003)	/	/
Outbreak-prosperity scenario	L_PRI	**0.564***** **(0.237, 0.892)**	0.086*** (0.042, 0.131)	/ /	/ /	/ /	/ /	/ /
	L_PRI_squ	**−0.005***** **(−0.008, −0.002)**	/ /	/ /	/ /	/ /	/ /	/ /
	L-SHI	/ /	/ /	0.632*** (0.280, 0.983)	0.077*** (0.034, 0.120)	/ /	/ /	**0.629***** **(0.268, 0.991)**
	L-SHI_squ	/ /	/ /	−0.006*** (−0.010, −0.002)	/ /	/ /	/ /	**−0.006***** **(−0.010, −0.002)**
	L-ESI	/ /	/ /	/ /	/ /	0.085 (−0.022, 0.193)	0.039** (0.007, 0.071)	**0.023** **(−0.015, 0.061)**
	L-ESI_squ	/ /	/ /	/ /	/ /	−0.0004 (−0.001, 0.0005)	/ /	/ /
Suppression-stagnancy scenario	L_PRI	**0.205*** **(−0.003, 0.412)**	−0.019 (−0.051, 0.014)	/ /	/ /	/ /	/ /	/ /
	L_PRI_squ	**−0.003**** **(−0.005, −0.0002)**	/ /	/ /	/ /	/ /	/ /	/ /
	L-SHI	/ /	/ /	0.219* (−0.007, 0.444)	−0.018 (−0.051, 0.016)	/ /	/ /	**0.245**** **(0.014, 0.477)**
	L-SHI_squ	/ /	/ /	−0.003** (−0.005, −0.0001)	/ /	/ /	/ /	**−0.003**** **(−0.006, −0.0002)**
	L-ESI	/ /	/ /	/ /	/ /	0.051 (−0.024, 0.125)	−0.019 (−0.048, 0.011)	**−0.0156** **(−0.041, 0.010)**
	L-ESI_squ	/ /	/ /	/ /	/ /	−0.001* (−0.001, 0.0001)	/ /	/ /
	AIC **BIC**	**328.9147** **392.4162**	**347.6178** **402.0476**	**336.7429** **400.2444**	**355.4985** **409.9284**	**369.6402** **433.1417**	**368.5781** **423.008**	**336.877** **409.4501**

AIC, Akaike Information Criterion; BIC, Bayesian Information Criterion; CI, confidence interval.
*$P < 0.1$; **$P < 0.05$; ***$P < 0.01$.
Model Total focused on the impact of overall policy response index (with the squared terms). Model Subtype-stringency/health policies and Model Subtype-economic-policies included stringency/health policy index (with its squared terms) and economic support policy index (with its squared terms), respectively. Models For reference were provided without squared terms. Model Subtype was the comprehensive model with the policy mix. L_PRI was policy response index. L-SHI and L-ESI referred to stringency/health policy index and economic support policy index. These Models all controlled for covariates. Bold models fitted the data better and were used to estimated probabilities for health-economic scenarios.

expansion in Q1 2021 along with a reported case count of more than 10 million people, resulting in the outbreak-prosperity scenario. In addition, Indonesia was in suppression-stagnancy in Q4 2020. The blockade for virus control kept economic growth low for a significant amount of time, with imports and exports greatly reduced (Fig. 1C and D). It was worth noting that the outbreak-stagnancy scenario logically followed from the coupled relationship, such as the increase in the number of confirmed cases from Q2 to Q3 of 2020 after the slow growth of the Indian economy, which brought India from the suppression-stagnancy to the outbreak-stagnancy scenario (Fig. 1A and B).

Associations between policy response and health-economic scenarios

Since Q1 2020, all countries adopted either strong or weak policies (for more details, see Figs. S2 and S3 in the supplementary material).

Table 1 shows the results of multinomial logistic regression with random effects. From Model Total, PRI and its squared terms were found to be statistically significant, and the coefficients of squared term were less than 0. The coefficients of PRI and its squared terms were 0.970 (0.271, 1.668) and −0.009 (−0.015, −0.003) for suppression-prosperity scenario; 0.564 (0.237, 0.892) and −0.005 (−0.008, −0.002) for outbreak-prosperity scenario; and 0.205 (−0.003, 0.412) and −0.003 (−0.005, −0.0002) for suppression-stagnancy scenario, respectively. Because AIC and BIC of Model Total were lower than the reference, the quadratic form may fit the data better. The results validated that the effects of policy response on health-economic scenarios were marginal diminishing. Inverse U-shapes can be seen in Fig. S4 in the supplementary material.

Stringency/health policies and economic support policies were analysed in Model Subtypes separately (Table 1). According to comparisons of AIC, BIC and the significance of the squared terms among these models, the effects of SHI were found to be inverted U-shaped (AIC and BIC in Model Subtype-stringency/health policies were lower than the reference, and the coefficients of the squared terms were mostly significant), whereas the effects of ESI were linear (AIC and BIC in Model Subtype-economic-policies were higher than the reference, and the coefficients of the squared terms were not significant). The regression was further performed on SHI, SHI's squared terms and ESI in Model Subtype. The coefficients of SHI and its squared terms remained significant, whereas the significance of ESI coefficients became lower.

For sensitivity analyses, a fixed effects multinomial logistic regression was first performed for comparison (see Table S7 in the supplementary material). Inverted U-shaped relationships continued to exist and were mainly significant; hence, the substantive findings remained unaltered. According to the results of Hausman tests, random effects models were shown as main analyses (see Table S8 in the supplementary material). Second, a logistic regression was performed after recoding the outcomes by setting the outbreak-stagnancy scenario and suppression-prosperity scenario as the base category separately. Analyses showed that most of the effects of policies followed the same patterns (see Table S9 in the supplementary material).

Estimated curves of policy response with health-economic scenarios

Model Total and Model Subtype (best-fitted) were used to estimate probabilities for health-economic scenarios (see Fig. 2). The outbreak-stagnancy scenario was almost certain when there was no policy at all. With increasing policy response, the probabilities of the other three scenarios increased. At medium-low intensity (around 30 score), the suppression-stagnancy scenario peaked for 42.80% (22.79%, 62.82%). A higher intensity of late policy response was likely to lead to outbreak-prosperity or suppression-prosperity scenarios. At medium-high intensity (about 57 score), the probability of suppression-prosperity reached the highest for 40.27% (25.14%, 55.40%); and outbreak-prosperity scenario reached the peak for 43.48% (27.40%, 59.56%) at the turning point of 66 score (see Table S10 in the supplementary material). The likelihood of these three eventualities decreased considerably when the policy intensity rose above a 70-score threshold.

Separately, the curves of suppression-prosperity, outbreak-prosperity and suppression-stagnancy scenarios corresponded to three different inverted U-shapes, which could be described as 'leptokurtic', 'bimodal' and 'long-tailed' (Fig. 2B–D). A leptokurtic shape indicated that the curvature of the policy response curve was larger, and marginal effects were more distinct and sensitive in the suppression-prosperity scenario. The bimodal shape demonstrated that the probability curve of the outbreak-prosperity scenario had a length of 'plateau' after increasing to a certain level and showed signs of relatively obvious twin-peak characteristics. The probability of the suppression-stagnancy scenario swiftly reached its peak and then rapidly declined with the increase of policy intensity, as it was long-tailed.

The three-dimensional figures of stringency/health and economic policy mix are shown in Fig. 2E–G. It was shown that moderate intensity of stringency/health policies and high intensity of economic support policies were the optimal policy mix. Positive incentives from SHI attained maximum at medium policy intensity. Yet, the impacts from ESI continued while enhancing policies.

Discussion

Data from OECD and BRIICS countries were used to analyse the impacts of policy response on the pandemic and the economy. The results showed that (1) both economic development and epidemic suppression might be achieved, namely, decoupled, with the intervention of policy response; (2) the impacts of policy response were lagged and showed an inverted U-shape, meaning that the benefits of policy response on the health-economic system would be overweighed by the incremental distortive effects after the turning point; (3) the effects of stringency/health policies resembled inverted U-shapes compared with economic support policies; and (4) estimated curves of suppression-prosperity, outbreak-prosperity and suppression-stagnancy scenarios presented 'leptokurtic', 'bimodal' and 'long-tailed' characteristics. These findings might help to keep the epidemic controlled while not sacrificing economy recovery in the early pandemic period.

It was often assumed that stimulating the economy and controlling the pandemic were in conflict.[35] However, the current analyses based on OECD and BRIICS countries offered a different story. This study emphasised that the dual objective, by decoupling the epidemic and the economy, was more important when facing complex, uncertain and ambiguous risks. It was possible to keep the epidemic controlled while not sacrificing the recovery of the economy. In addition, this study focused on the perspective of policy intensity and put forward the inverted U-shaped pattern of policy effects, indicating that it was advisable to adopt medium-intensity (around 57 score) public policies in the early COVID-19 period (see Table S10 in the supplementary material). This U-shaped pattern is consistent with several previous studies on pandemic policy analyses.[9,36] In total, 20 different policies and 38 countries were included in the current analyses, which enhanced the general significance of the research results. The study findings are complementary to the previous literature.

From a theoretical perspective, the inverted U-shaped curves of stringency/health policies may be caused by the interaction of two latent but opposite functions.[37] Because these functions are asymmetric in two aspects, an inverted U emerges. The first stems from the asymmetry between the effects of blockade policies on health and economic activities. Lockdowns are detrimental to the economy but are beneficial for viral containment. However, panic and irrational behaviour responses as a result of lockdowns often result in extensive damage to economic systems, which is greater than the protective effect provided by lockdowns to susceptible people.[38] The second stems from the asymmetry between the effects of health support policies on the contact rate and the probability of infection. Scholars often believe that health support policies can separate the contact rate and the probability of infection, which should both be dependent on social mobility.[39,40] In terms of the probability of infection, the initial effect of health policies is obvious, but higher levels of intensity may not continue to decrease this variable.[5,41] For example, more extensive tests sometimes have no greater effect than tests for high-risk populations. In terms of the contact rate, the incremental policies may

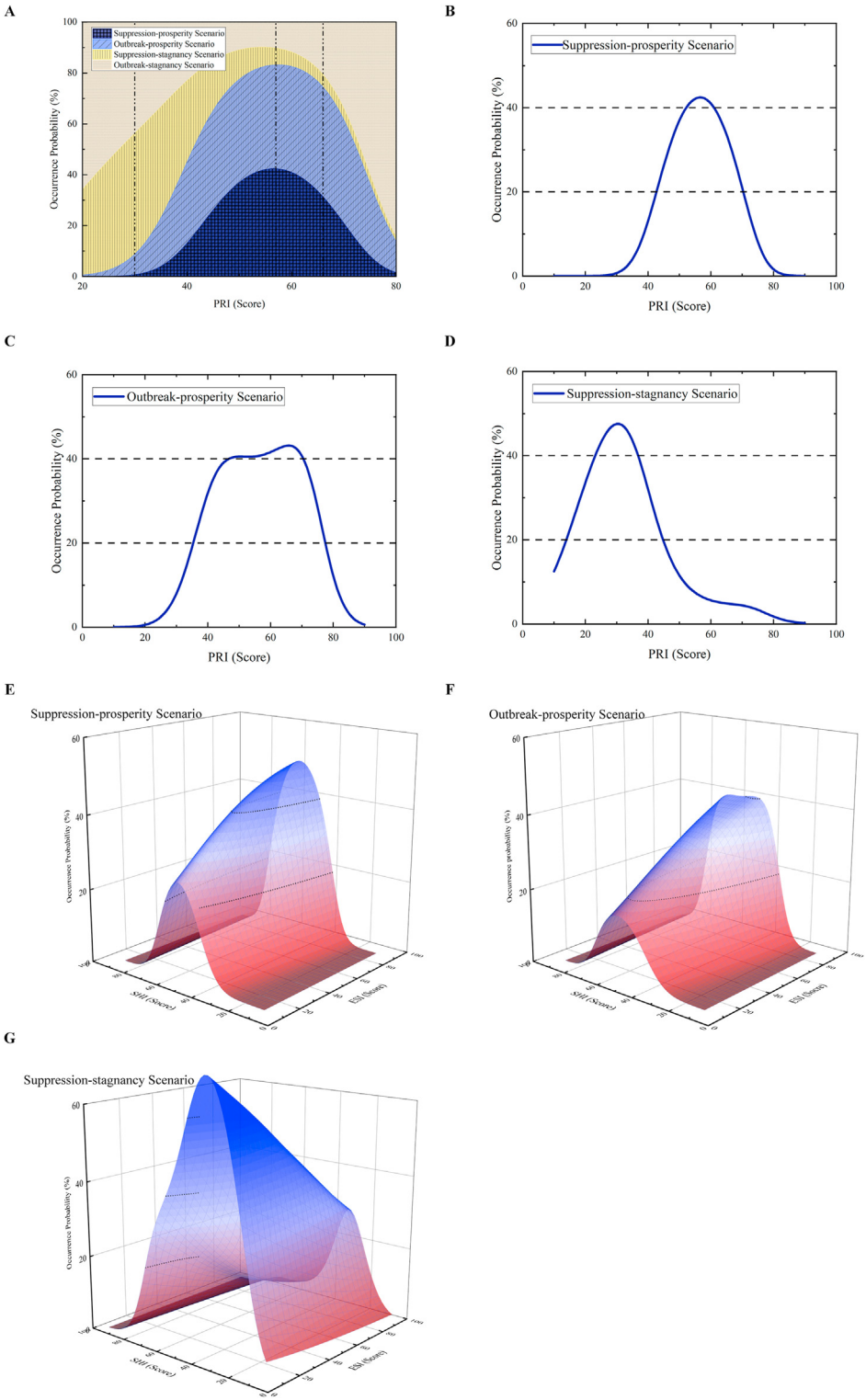

Fig. 2. Estimated curves of policy response and marginal effects. Panel A indicates the probabilities of four scenarios; Panels B–D represents the probability curves of three interest scenarios as PRI changed; Panels E–G were probability curves under the impact of stringency/health policies and economic support policies. PRI, policy response index.

make people overtrust the government's ability and become more relaxed about self-protection and movement restrictions, both psychologically and behaviourally.[42] When the promotion of the contact rate outpaces the inhibition of the probability of infection, negative incentives outweigh positive ones. From these two perspectives, the effects of stringency/health policies show inverted U-shapes.

This study identified 'leptokurtic', 'bimodal' and 'long-tailed' inverted U-shapes for the scenarios of interest. In probability theory, leptokurtic shapes are curves with larger positive kurtosis,

bimodal shapes refer to curves with two peaks that indicate two local maximums and long-tailed shapes have heavier tails than the typical.[43] Disparities among the three scenarios were caused by the dissimilar policy thresholds and unequal widths of effective policy intervals. The probabilities of scenarios were affected by each other, and the dominant scenario was different under different policy intensity. Suppression-prosperity scenario was the optimal outcome, but the most difficult to achieve during the pandemic. The kurtosis of the curve of suppression-prosperity scenario was relatively large, where even a slight intensity score deviation from the turning point could result in a considerable decrease in probability. Moreover, the effective policy interval for the suppression-prosperity scenario was smaller than that in the outbreak-prosperity scenario. Hence, the effective interval of policy intensity for suppression-prosperity scenario was very narrow. In a sense, 'moderate' policy measures are the key to good governance. To be specific, too-strong or too-weak policy responses were detrimental to achieve the suppression-prosperity scenario, and precise intensity of policy implementation was essential.[44]

Although this study has demonstrated the effectiveness of some policy response, the possibility of action is limited by the constantly changing virus, and thus, policymakers must be prudent. Policy instruments are more likely to be implemented when institutional and social infrastructures support management actions.[45] However, in some decentralised systems, governments face higher thresholds to justify intervention.[12] In addition, China's recent experience revealed that the ever-increasing 'ultra-long isolation' did not benefit the dual objective and increased the cost of controlling the epidemic. Policymakers should be cautious to maximise the effectiveness of policy implementation by ensuring that the reaction offers positive rather than negative incentives.

This study has some limitations. First, treating only the strength of policy response might fail to consider some crucial concerns pertaining to policy contents. In addition, to perform country-based comparison, the set outcomes might overlook some contexts in this study. For example, over the study period, different countries were dealing with different COVID-19 variants and conducted testing at different rates. Quarterly variations could also make the current data less comparable. Moreover, the inverted U-shaped curves identified did not emphasise predictive functions but rather illustrated the tendency for policy intensity to affect health-economic scenarios. In addition, the temporality of the data in the early pandemic phase also limited the study. Only the early stages of the pandemic were considered in the assessments that were given in the present study. Future research could consider the contents of policy instruments, as well as the difference of testing rates and viral variants.

In conclusion, it was possible to achieve the dual objective of economic growth and epidemic control simultaneously, and the effects of policy response were shaped like an inverse U. These findings provide a new perspective for balancing the economy with public health during the early stages of pandemics.

Author statements

Ethical approval

The study used publicly available and country-level data with no identifiable personal information; thus, ethical approval was not required.

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Competing interests

None declared.

Data availability

The data underlying this article are available in WHO, OECD, OxCGRT and Wind databases.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.puhe.2023.01.012.

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